# Immunodeficiency 133 With Ectodermal Dysplasia With Or Without Peripheral Neuropathy (IMD133): Comprehensive Disease Report

**Disease:** Immunodeficiency 133 With Ectodermal Dysplasia With Or Without Peripheral Neuropathy (IMD133) **MONDO ID:** MONDO:0979570 · **OMIM:** #621254 · **Causal gene: ITPR3** (IP3R3; OMIM 147267; HGNC:6182; chromosome 6p21) **Synonyms:** IMD133; ITPR3-related combined immunodeficiency; IP3R3 dominant-negative combined immunodeficiency

## Summary

Immunodeficiency 133 with ectodermal dysplasia with or without peripheral neuropathy (IMD133) is an ultra-rare multisystem Mendelian disorder caused by variants in **ITPR3**, the gene encoding **IP3R3 (inositol 1,4,5-trisphosphate receptor type 3)** — a subunit of the tetrameric ER calcium-release channel. The disease is driven predominantly by **de novo heterozygous dominant-negative missense variants** (and, less commonly, biallelic/recessive variants) that impair channel function. A single mutant subunit, incorporated into the four-subunit (tetrameric) channel, poisons the whole complex. This is **not** a haploinsufficiency mechanism. The resulting defect depletes endoplasmic reticulum (ER) $Ca^{2+}$ stores and blunts **store-operated $Ca^{2+}$ entry (SOCE)** in T cells, crippling the NFAT- and NF-κB-dependent signaling required for normal T-cell development and activation (PMID: 39560673).

The clinical picture is a **combined immunodeficiency (CID)** — recurrent bacterial, viral, and fungal infections beginning in the first months to years of life, T-cell lymphopenia, naive T-cell depletion, hypogammaglobulinemia, and immune dysregulation (immune thrombocytopenia, autoimmune hemolytic anemia, lymphoproliferation). Susceptibility to EBV, VZV, and HPV is prominent, and EBV-driven lymphoma and hemophagocytic lymphohistiocytosis (HLH) are major life-threatening complications. Because IP3R-dependent $Ca^{2+}$ signaling is also essential for ectodermal epithelia (notably enamel-forming ameloblasts) and peripheral nerve, affected

individuals variably show **ectodermal dysplasia** (conical/peg-shaped teeth, hypodontia, microdontia, sparse fine hair, hypohidrosis, dysplastic nails) and **demyelinating/intermediate peripheral neuropathy** (allelic with Charcot-Marie-Tooth disease type 1J, CMT1J).

The gene displays a striking **allelic/phenotypic spectrum**: the same gene causes dominant CMT neuropathy (CMT1J) and dominant-negative combined immunodeficiency with ectodermal dysplasia, and the phenotypes can overlap. Fewer than ~20 genetically confirmed IMD133 patients have been reported worldwide. Diagnosis rests on **trio whole-exome/whole-genome sequencing** (to catch de novo variants) or a primary-immunodeficiency gene panel including ITPR3, supported by functional $Ca^{2+}$/SOCE assays. **Hematopoietic stem-cell transplantation (HSCT)** can correct the immune defect, but no targeted or gene therapy currently exists.

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## Key Findings

### Finding 1 — IMD133 is caused by de novo dominant-negative missense variants in ITPR3 (IP3R3)

Blanco et al. (2025, *J Exp Med*; PMID: 39560673) reported **five unrelated patients with de novo missense variants in ITPR3** presenting with combined immunodeficiency, abnormal T-cell $Ca^{2+}$ homeostasis, incompletely penetrant ectodermal dysplasia, and multisystem disease. The authors state: *"We report five unrelated patients with de novo missense variants in ITPR3, encoding a subunit of the inositol 1,4,5-trisphosphate receptor (IP3R)."*

Critically, pathogenicity is **not due to haploinsufficiency**. Instead, *"ITPR3 protein variants interfere with IP3R channel function leading to depletion of ER Ca2+ stores and blunted SOCE in T cells."* Because IP3R3 assembles as a tetramer, one defective subunit disrupts the whole channel — the textbook definition of a **dominant-negative** mechanism. This establishes both the causal gene and the molecular class of lesion.

### Finding 2 — ITPR3 shows an allelic/phenotypic spectrum spanning dominant CMT neuropathy and dominant-negative combined immunodeficiency

ITPR3 variants cause a continuum of disease. On the neuropathy end: - Rönkkö et al. (2020; PMID: 32949214) first proved ITPR3 is a CMT gene: *"Exome sequencing identified a heterozygous ITPR3 p.Val615Met variant segregating with the disease"* (AD family, demyelinating + axonal,

onset ~age 30) and a de novo p.Arg2524Cys (onset age 4, severe). - Beijer et al. (2024; PMID: 38938188) identified a recurrent hotspot: *"A recurrent heterozygous missense variant p.Thr1424Met ... was present in 33 affected individuals from nine unrelated families"* — all with slow nerve conduction velocities (CMT1). - Cabello-Murgui et al. (2024; PMID: 39287469) reported *"The c.4271C > T ITPR3 variant causes a late onset CMT that can be considered an intermediate CMT"* — initially misdiagnosed as CIDP.

Peripheral neuropathy, immunodeficiency, and tooth abnormalities can co-occur in the same individuals, which is why the disease name carries the "with or without peripheral neuropathy" qualifier.

## Finding 3 — A spontaneous canine ITPR3-null model recapitulates CMT1J with ectodermal (enamel) involvement

Hytönen et al. (2025; PMID: 39804930) identified *"a homozygous nonsense variant in the ITPR3 gene in Lancashire Heeler dogs, presenting with a severe developmental enamel defect and reduced nerve conduction velocity."* At the protein level, *"the nonsense variant in ITPR3 led to a complete absence of full-length IP3R3 protein. Unexpectedly, the protein levels of IP3R1 and IP3R2 were also markedly decreased, suggesting co-regulation,"* and IP3R-mediated $Ca^{2+}$ flux upon GPCR stimulation was reduced. This natural model links ITPR3 loss to both the ectodermal (amelogenesis imperfecta) and neuropathic phenotypes of the human disease.

## Finding 4 — IMD133 = OMIM #621254; a multisystem autosomal-dominant disorder

OMIM #621254 defines IMD133 as caused by heterozygous mutation in ITPR3 (OMIM 147267; HGNC:6182) at 6p21. Reported de novo heterozygous missense variants across the defining cohorts include **A196T, I2506N, R2524C (recurrent), and R2524H** (Blanco et al. 2025; PMID: 39560673) and a recurrent **R2524C** in four patients (Molitor et al. 2024, *Sci Adv*; PMID: 39270020). Core features: combined immunodeficiency/CVID with recurrent infection in the first months-to-years of life, T-cell lymphopenia and dysfunction, variable ectodermal dysplasia, ± peripheral neuropathy, immune dysregulation, poor growth/short stature, and developmental delay — *"incompletely penetrant ectodermal dysplasia, and multisystem disease."*

## Finding 5 — T-cell phenotype: lymphopenia, naive depletion, restricted TCR, impaired NFAT/NF-κB and mitochondrial/metabolic downregulation

Blanco et al. (2025; PMID: 39560673) describe *"defects in late stages of thymic T cell development, and impaired function of peripheral T cells, including inadequate NF-κB- and NFAT-mediated, proliferative, and metabolic responses to activation."* Molitor et al. (2024; PMID: 39270020) in R2524C patients found $CD4^+$ lymphopenia, quasi-absence of naive $CD4^+$ and $CD8^+$ cells, increased memory cells, a distinct/restricted TCR repertoire, defective calcium homeostasis, and mitochondrial malfunction, with decreased proliferation and downregulation of metabolic and mitochondrial pathways.

## Finding 6 — Clinical course, complications, and treatment

Per OMIM #621254 and Blanco et al. (2025; PMID: 39560673), patients present in the first months-to-years of life with recurrent bacterial, viral, and fungal infections (*"The patients presented with CID, abnormal T cell Ca2+ homeostasis, incompletely penetrant ectodermal dysplasia, and multisystem disease"*), hypogammaglobulinemia/B-cell abnormalities, and immune dysregulation (immune thrombocytopenia, autoimmune hemolytic anemia, lymphoproliferation). Of the 5 Blanco patients: 2 underwent **HSCT** (ages 9 and 3); 2 developed **EBV-driven lymphomas** (ages 22 and 12) requiring chemotherapy then HSCT; the 5th remained clinically well on antibiotic prophylaxis at age 18. The Molitor cohort additionally reported bone-marrow failure in some patients.

## Finding 7 — IP3R3 protein: a tetrameric ER $Ca^{2+}$-release channel poisoned by dominant-negative variants

ITPR3 encodes IP3R3, a ~2,671-amino-acid subunit that assembles as a homo/heterotetramer forming an ER-membrane $Ca^{2+}$ channel, with an N-terminal IP3 ligand-binding (and suppressor) domain and a C-terminal transmembrane channel/pore domain (OMIM 147267). Blanco et al. describe the subunit *"which forms a Ca2+ channel in the endoplasmic reticulum (ER) membrane responsible for the release of ER Ca2+ required to trigger SOCE, and for Ca2+ transfer to other organelles."* Because four subunits co-assemble, a single mutant subunit disrupts the tetramer — the basis for the dominant-negative (not haploinsufficient) mechanism. IP3R3 also localizes to **mitochondria-associated ER membranes (MAMs)**, where it transfers $Ca^{2+}$ to mitochondria to sustain bioenergetics: *"the MAM is a critical cell-signaling junction whereby Bcl-xL dynamically interacts with IP3R3 to coordinate mitochondrial Ca(2+) transfer and alters cellular metabolism in order to increase the cells' bioenergetic capacity"* (PMID: 27155879).

## Finding 8 — HPO phenotype spectrum with frequencies

Aggregated across three reports (~11 patients; PMID: 39560673, PMID: 39270020, PMID: 36302985), see the phenotype table below.

## Finding 9 — Ectodermal dysplasia branch: IP3R-dependent ER $Ca^{2+}$ release feeds SOCE required for enamel/ectodermal development

In dental enamel cells, Nurbaeva et al. (2015; PMID: 26515404) found *"inositol 1,4,5-trisphosphate receptor (IP3R) but not ryanodine receptor (RyR) expression was high in enamel cells suggesting that IP3Rs are the main ER Ca(2+) release mechanism."* SOCE-driven $Ca^{2+}$ signals then drive enamel matrix protein gene expression: *"Ca2+ signals enhance the expression of EMPs"* (PMID: 26232387). Patients with CRAC-channel (STIM1/ORAI1) mutations develop amelogenesis imperfecta, and in IMD133 ectodermal dysplasia occurs in 4/5 patients, while the canine ITPR3-null model shows severe amelogenesis imperfecta (PMID: 39804930).

## Finding 10 — Inheritance heterogeneity and severe EBV-driven HLH

The first ITPR3 immunodeficiency report (Neumann/Ehl et al. 2022, *Cell Mol Immunol*; PMID: 36302985) described two unrelated patients with **compound heterozygous (biallelic/recessive)** ITPR3 variants — a 12-year-old with CID (HSCT at age 6) and a 36-year-old with CVID — establishing IP3R defects as a new class of inborn errors of immunity. Subsequent cohorts (Molitor 2024; Blanco 2025) describe **de novo heterozygous dominant-negative** variants. A 2025 report (PMID: 41019080) describes two CID patients with ITPR3 mutations presenting with life-threatening severe EBV-associated HLH. Blanco et al. frame the biology: *"The importance of calcium (Ca2+) as a second messenger in T cell signaling is exemplified by genetic deficiencies of STIM1 and ORAI1, which abolish store-operated Ca2+ entry (SOCE) resulting in combined immunodeficiency (CID)."*

## Finding 11 — Model organisms

Mouse (NCBITaxon:10090; *Itpr3* NCBI Gene 16440): Futatsugi et al. (2005, *Science*; PMID: 16195467) — *"We show exocrine dysfunction in IP3R2 and IP3R3 double knock-out mice, which caused difficulties in nutrient digestion,"* and *"Despite a normal caloric intake, the double mutants were hypoglycemic and lean."* Single Itpr3-KO mice have mild phenotypes (paralog redundancy). Dog (NCBITaxon:9615): spontaneous ITPR3-null Lancashire Heeler model (PMID:

39804930). Cellular models: Jurkat T-cell lines engineered with R2524C reproduce dominant-negative defective $Ca^{2+}$ flux (PMID: 39270020); patient fibroblasts and T cells show depleted ER stores and blunted SOCE.

## Finding 12 — Diagnostics and epidemiology

Ultra-rare: fewer than ~20 genetically confirmed patients worldwide; no Orphanet/GBD prevalence estimate. Pathogenic alleles are absent/ultra-rare in gnomAD (CMT hotspot p.Thr1424Met *"gnomAD v4 allele frequency 1.76 × 10-6"*; PMID: 38938188). Diagnosis via trio WES/WGS or a PID gene panel including ITPR3. Supportive labs: T-cell lymphopenia (*"significant T cell lymphopenia, defects in late stages of thymic T cell development"*), low $CD8^+$/naive T cells, low/abnormal TRECs (potentially detectable on newborn SCID screening), hypogammaglobulinemia/low IgM, restricted TCR, impaired proliferation. Functional confirmation by Fura-2/Fluo-4 $Ca^{2+}$ flux and SOCE assays after thapsigargin or TCR stimulation.

## Phenotype Spectrum (HPO terms and frequencies)

| Domain | Phenotype (HPO) | Frequency (aggregated) |
|---|---|---|
| Immune (core) | Combined immunodeficiency (HP:0005387) | 5/5 |
| Immune | Decreased total T cell count (HP:0005403) | 15/15 |
| Immune | Decreased total lymphocyte count (HP:0001888) | 10/10 |
| Immune | Decreased $CD8^+$ T cell proportion (HP:0005415) | 8/9 |
| Immune | Decreased naive $CD4^+$ (HP:0410378) / $CD8^+$ (HP:0410377) | 3/3 |
| Immune | Decreased circulating IgM (HP:0002850) | 4/4 |
| Infection | Pneumonia (HP:0002090) | 4/4 |
| Infection | Bronchiectasis (HP:0002110) | 2/2 |
| Infection | Severe VZV (HP:0032170) | 2/2 |
| Infection | Molluscum contagiosum (HP:0032163) | 2/2 |
| Infection | Persistent EBV viremia (HP:0020072); Warts (HP:0200043) | reported |
| Dysregulation | Autoimmune thrombocytopenia (HP:0001973) | 3/4 |
| Malignancy | Lymphoma (HP:0002665); B-cell (HP:0012191); Hodgkin (HP:0012189) | reported |
| Ectodermal | Ectodermal dysplasia (HP:0000968) | 4/5 |
| Ectodermal | Fine hair (HP:0002213) | 4/5 |
| Ectodermal | Peg-shaped maxillary lateral incisors (HP:0006342) | 3/4 |
| Ectodermal | Conical incisor (HP:0011065); Microdontia (HP:0000691) | reported |
| Ectodermal | Hypodontia (HP:0000668) | 2/4 |
| Ectodermal | Hypohidrosis (HP:0000966) | 1/5 |
| Ectodermal | Sparse eyebrow (HP:0045075) | 2/2 |
| Neuro | Demyelinating peripheral neuropathy (HP:0007108); Polyneuropathy (HP:0001271) | reported |

| Domain | Phenotype (HPO) | Frequency (aggregated) |
|---|---|---|
| Neuro | Global developmental delay (HP:0001263); Intellectual disability (HP:0001249) | reported |
| Neuro | Delayed speech/language (HP:0000750) | 5/5 |
| Neuro | Motor delay (HP:0001270) | 4/4 |
| Neuro | Pes cavus (HP:0001761); Hearing impairment (HP:0000365) | 2/4 (hearing) |
| Growth | Growth delay (HP:0001510) | 4/7 |
| Onset | Infantile onset (HP:0003593) | 6/11 |

## Mechanistic Model / Causal Chain

The pathophysiology is organized as a branched causal chain from the initiating mutation to the multisystem clinical manifestations. Steps marked *(inferred)* are mechanistically plausible but not directly demonstrated in IMD133 patients.




**Upstream vs downstream:** Steps 1–3 (mutation → dominant-negative channel → ER $Ca^{2+}$ depletion) are the shared upstream lesion. The four branches are downstream, tissue-specific consequences reflecting where IP3R3-dependent $Ca^{2+}$ signaling is non-redundant.

**Cell types and processes (ontology suggestions):** - **Biological processes (GO):** release of sequestered calcium ion into cytosol (GO:0051209); store-operated calcium entry (GO:0002115); calcium-mediated signaling (GO:0019722); T cell activation (GO:0042110); positive regulation of NF-κB / NFAT transcription factor activity. - **Cellular components (GO):** endoplasmic reticulum membrane (GO:0005789); inositol 1,4,5-trisphosphate receptor complex; mitochondria-associated ER membrane (MAM). - **Cell types (CL):** T cell (CL:0000084), naive T cell (CL:0000898), thymocyte (CL:0000893), ameloblast (CL:0000059), Schwann cell (CL:0002573), B cell (CL:0000236). - **Anatomy (UBERON):** thymus (UBERON:0002370), tooth enamel/ameloblast layer, peripheral nerve (UBERON:0001021), skin/hair follicle, nail. - **Chemical (CHEBI):** calcium(2+) (CHEBI:29108), inositol 1,4,5-trisphosphate (CHEBI:16595).

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## Section-by-Section Content

### 1. Disease Information

IMD133 is an inborn error of immunity with ectodermal and (variable) neuropathic features caused by ITPR3/IP3R3 dysfunction. **Identifiers:** OMIM #621254; MONDO:0979570; gene ITPR3 (OMIM 147267, HGNC:6182). Orphanet/ICD-10/ICD-11/MeSH codes are not yet assigned (ultra-rare, recently delineated). **Synonyms:** IMD133; ITPR3-related combined immunodeficiency; IP3R3 dominant-negative combined immunodeficiency. Information is derived from **individual-patient case series** (not aggregated disease-level registries).

### 2. Etiology

**Causal factor:** monogenic — pathogenic variants in ITPR3. **Genetic risk:** de novo dominant-negative missense (A196T, I2506N, R2524C, R2524H) and, less commonly, biallelic/compound-heterozygous variants. No established environmental risk or protective factors; no modifier genes defined. **Gene-environment interaction:** infectious exposures (EBV, VZV, HPV) unmask/drive the most severe complications (lymphoma, HLH) on the background of the T-cell defect — a gene × pathogen interaction rather than a classical GxE.

## 3. Phenotypes

See the phenotype table above. Onset is typically infantile; severity is variable (mild/conservatively managed to lethal lymphoma/HLH); course is chronic and multisystem. Quality-of-life impact is substantial (recurrent infections, transplant, malignancy risk, dental/dermatologic features, possible neuropathy and developmental delay); no formal QOL instrument data exist for this ultra-rare disease.

## 4. Genetic/Molecular Information

**Causal gene:** ITPR3 (6p21). **Variant types:** missense (dominant-negative, de novo) predominate; nonsense (recessive, canine model). **Classification:** reported disease variants are pathogenic/likely pathogenic per ACMG (de novo occurrence, functional evidence). **Allele frequency:** absent/ultra-rare in gnomAD. **Origin:** germline (de novo or inherited). **Functional consequence:** dominant-negative channel dysfunction (ER $Ca^{2+}$ depletion, blunted SOCE) — not haploinsufficiency. No epigenetic or chromosomal-abnormality mechanism; modifier genes unknown.

## 5. Environmental Information

No causal toxins, radiation, or occupational exposures. **Infectious agents** act as triggers/complicating factors: EBV (lymphoma, HLH), VZV (severe/disseminated), HPV (warts), molluscum contagiosum, plus general bacterial/fungal pathogens.

## 6. Mechanism / Pathophysiology

See the causal-chain diagram and ontology suggestions above. Core pathway: IP3/$Ca^{2+}$ signaling → SOCE (ORAI/STIM) → calcineurin-NFAT and NF-κB (KEGG calcium signaling; Reactome SOCE). Downstream processes: impaired T-cell development/activation, mitochondrial/metabolic reprogramming (MAM $Ca^{2+}$ transfer), enamel mineralization, and peripheral myelination.

## 7. Anatomical Structures Affected

**Primary:** immune system (thymus, T lymphocytes; UBERON:0002370). **Ectodermal derivatives:** teeth/enamel (ameloblasts), hair follicles, skin sweat glands, nails. **Nervous system:** peripheral nerves (Schwann cells/axons). **Secondary/systemic:** growth (short stature), hematologic (cytopenias, bone-marrow failure), lymphoid organs (lymphoproliferation/lymphoma). **Subcellular:** endoplasmic reticulum membrane and MAMs. Lateralization: bilateral/systemic.

## 8. Temporal Development

**Onset:** congenital/infantile for immune and ectodermal features; neuropathy onset variable (childhood to adult, allelic CMT range). **Progression:** chronic, lifelong; punctuated by acute, potentially fatal events (severe infection, EBV lymphoma/HLH). No spontaneous remission; HSCT is curative for the immune component. Critical window: early life, before severe infection/malignancy and with the opportunity for timely HSCT.

## 9. Inheritance and Population

**Inheritance:** predominantly autosomal dominant (de novo); also autosomal recessive (biallelic). **Penetrance:** immune phenotype highly penetrant; ectodermal dysplasia incompletely penetrant. **Expressivity:** highly variable. **Epidemiology:** ultra-rare, <~20 confirmed patients worldwide; no prevalence/incidence estimate. No founder effect, defined carrier frequency, or sex predilection established.

## 10. Diagnostics

**Genetic:** trio WES/WGS (de novo detection) or PID gene panel including ITPR3 (Genomics England PanelApp). **Immunologic labs:** T-cell lymphopenia; low $CD8^+$/naive T cells; low/abnormal TRECs (newborn SCID screen may flag); hypogammaglobulinemia/low IgM; restricted TCR; impaired PHA/anti-CD3/CD28 proliferation. **Functional confirmation:** Fura-2/Fluo-4 $Ca^{2+}$ flux and SOCE assays (thapsigargin or TCR stimulation) showing depleted ER stores and blunted SOCE. **Nerve conduction studies** for neuropathy. **Differential diagnosis:** STIM1/ORAI1 CRAC-channelopathies, other CID/CVID, other ectodermal dysplasias, CMT of other genetic causes, CIDP (neuropathy can mimic).

## 11. Outcome/Prognosis

Guarded; depends on infection control, malignancy, and HSCT success. Major mortality drivers: severe/opportunistic infection, EBV-driven lymphoma and HLH, bone-marrow failure. HSCT can be curative for the immunologic defect but carries transplant-related risk and does not reverse established ectodermal or neuropathic features. No large survival dataset exists. Prognostic factors: variant severity, infection/EBV burden, timeliness of transplant.

## 12. Treatment

**Supportive/pharmacologic:** antimicrobial prophylaxis; immunoglobulin replacement (IVIG/SCIG) for hypogammaglobulinemia; antivirals for EBV/VZV; management of autoimmune cytopenias. **Definitive: hematopoietic stem-cell transplantation (HSCT; NCIT C15431)** corrects the immune defect. **Malignancy:** chemotherapy for EBV-driven lymphoma, often followed by HSCT; HLH-directed therapy for HLH. No targeted, RNA-based, or gene therapy currently exists. **NCIT suggestions:** Hematopoietic Cell Transplantation (C15431), Immunoglobulin Therapy (C603), Antimicrobial Prophylaxis.

## 13. Prevention

No primary prevention (de novo origin). **Secondary:** newborn SCID screening (low TRECs may detect T-cell lymphopenia early); early genetic diagnosis to enable prophylaxis and transplant planning. **Tertiary:** infection prophylaxis, EBV surveillance, dental/dermatologic care, neuropathy rehabilitation. **Genetic counseling:** mostly de novo (low recurrence) but germline mosaicism cannot be excluded; recessive families require carrier counseling.

## 14. Other Species / Natural Disease

**Dog** (*Canis lupus familiaris*, NCBITaxon:9615): spontaneous homozygous ITPR3 nonsense variant in Lancashire Heelers causes amelogenesis imperfecta + reduced nerve conduction velocity (CMT1J-like) — a natural veterinary model (PMID: 39804930). Orthologs: mouse *Itpr3* (NCBI Gene 16440). Evolutionarily conserved $Ca^{2+}$-channel function underlies cross-species phenotype overlap.

## 15. Model Organisms

- **Mouse:** *Itpr2/Itpr3* double-knockout (exocrine dysfunction, hypoglycemia, leanness, impaired growth; PMID: 16195467); single *Itpr3*-KO mild (paralog redundancy). Models the growth/metabolic branch; does not fully recapitulate the human dominant-negative immune phenotype.
- **Dog:** spontaneous ITPR3-null (above) — ectodermal + neuropathic recapitulation.
- **Cellular/in vitro:** Jurkat T cells expressing R2524C (dominant-negative $Ca^{2+}$ flux defect; PMID: 39270020); patient fibroblasts and primary T cells (depleted ER stores, blunted SOCE).
- **Resources:** MGI (mouse), OMIA (dog), Cellosaurus (Jurkat).

## Evidence Base

| PMID | Study (year) | Type | Contribution |
|---|---|---|---|
| 39560673 | Blanco et al. (2025) *J Exp Med* | Human clinical + functional | Defines IMD133: 5 patients, de novo dominant-negative ITPR3 variants, ER $Ca^{2+}$ depletion, blunted SOCE, T-cell defects |
| 39270020 | Molitor et al. (2024) *Sci Adv* | Human + Jurkat model | Recurrent R2524C; naive T-cell depletion, restricted TCR, mitochondrial dysfunction |
| 36302985 | Neumann/Ehl et al. (2022) *Cell Mol Immunol* | Human clinical | First ITPR3 immunodeficiency; biallelic/ recessive form; CID + CVID |
| 41019080 | (2025) *Front Immunol* | Human clinical | Severe EBV-driven HLH in ITPR3 CID |
| 32949214 | Rönkkö et al. (2020) | Human + functional | ITPR3 established as CMT gene (p.Val615Met, p.Arg2524Cys) |
| 38938188 | Beijer et al. (2024) | Human cohort | Recurrent p.Thr1424Met hotspot (33 individuals); gnomAD ultra-rarity |
| 39287469 | Cabello-Murgui et al. (2024) | Human clinical | Intermediate adult-onset CMT; CIDP misdiagnosis |
| 39804930 | Hytönen et al. (2025) | Canine model | ITPR3-null dog: amelogenesis imperfecta + neuropathy; IP3R paralog co-regulation |
| 16195467 | Futatsugi et al. (2005) *Science* | Mouse model | Itpr2/3 dKO: exocrine/metabolic/growth phenotype |
| 27155879 | (review) | Mechanistic | IP3R3-Bcl-xL MAM axis; ER→mito $Ca^{2+}$ and bioenergetics |
| 26515404 | Nurbaeva et al. (2015) | In vitro | IP3R is main ER $Ca^{2+}$ release in enamel cells |
| 26232387 | Nurbaeva et al. (2015) | In vitro | SOCE $Ca^{2+}$ signals drive enamel matrix protein genes |

## Limitations and Knowledge Gaps

1. **Tiny sample size.** Fewer than ~20 genetically confirmed patients; all frequency and penetrance figures derive from small case series and are not population-based.
2. **Neural branch is inferred.** The Schwann-cell/axon mechanism (Branch C) is extrapolated from allelic CMT studies, not directly demonstrated in IMD133 immunodeficiency patients.
3. **Ectodermal step 5b is inferred.** The link from reduced ameloblast $Ca^{2+}$ to decreased enamel-matrix-protein expression is supported by CRAC-channel and in vitro data, not by direct ITPR3-patient enamel transcriptomics.
4. **Genotype-phenotype correlation is unresolved.** Why some variants favor neuropathy (CMT1J) and others combined immunodeficiency — and why ectodermal dysplasia is incompletely penetrant — is not established.
5. **No validated biomarkers, QOL data, prognostic models, or formal epidemiology** (no Orphanet/GBD/ICD codes).
6. **No disease-specific mouse model** of the human dominant-negative immune phenotype; current mouse data model only the metabolic/growth branch.
7. **No targeted therapy.** HSCT is the only disease-modifying option and addresses only the immune component.

## Proposed Follow-up Experiments / Actions

1. **Knock-in mouse / zebrafish models** carrying human dominant-negative alleles (e.g., R2524C) to test immune, ectodermal, and neural recapitulation and genotype-phenotype links.
2. **Patient iPSC-derived systems:** iPSC→T cells, iPSC→ameloblast-like cells, and iPSC→Schwann cells to directly test each branch's $Ca^{2+}$/SOCE defect and transcriptional consequences.
3. **Structure-function mapping** of IP3R3 variants (cryo-EM/channel electrophysiology) to classify gain-of-function "leak" vs loss-of-function gating defects and correlate with phenotype severity.
4. **Pharmacologic rescue screens:** test whether SOCE potentiators, $Ca^{2+}$-modulating agents, or chaperone strategies restore T-cell $Ca^{2+}$ dynamics in patient cells.
5. **Natural-history registry** (international) to define penetrance, QOL, malignancy incidence, HSCT outcomes, and neuropathy progression.

6. **Newborn-screening analysis:** quantify TREC-based detectability to determine whether IMD133 could be caught on existing SCID panels.
7. **EBV-directed prevention/therapy evaluation** (surveillance, rituximab, EBV-specific CTLs) given the high lymphoma/HLH burden.

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*Report compiled from 13 confirmed findings and 26 reviewed papers across 5 investigation iterations. Evidence types are distinguished as human clinical, model organism, in vitro, or computational throughout.*

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